NPL Strategy Optimization: Efficient Frontier

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This project demonstrates how to use **Multi-Objective Optimization** to maximize debt recovery while minimizing operational costs in a Non-Performing Loan (NPL) portfolio.

🚀 The Solution

Using the **NSGA-II Genetic Algorithm**, the system evaluates thousands of potential strategy combinations to find the "Pareto-Optimal" set of actions (SMS, Calls, Legal).

🛠️ Tech Stack

- **Python** (Logic & Data Processing)
- **Platypus-opt** (Multi-Objective Optimization)
- **Streamlit** (Interactive Dashboard)
- **Pandas/NumPy** (Vectorized Calculations)

📊 Key Features

- **Dynamic Constraints:** Adjust budgets and legal capacity in real-time.
- **Efficient Frontier Visualization:** View the trade-off curve between cost and cure.

📈 Understanding the Frontier

- **The Curve:** Represents the "Most Efficient" path. Any point on this line is the maximum possible recovery for that specific spend.
- **The Flattening:** This represents Diminishing Marginal Returns. At a certain point, spending an extra $10k on collections yields less than $10k in recovery.

- **The "Legal Gap":** The gray dashed line shows the theoretical maximum recovery if legal capacity were infinite. The red line shows our current reality. The distance between them is the Opportunity Cost of our current operational constraints.

📺 Screenshots